

Name: Aijing Huang
CS6650: Scalable Distributed Systems – Assignment 3

Test Run Successfully: all 500,000 messages broadcasted successfully

====Load test completed=====

=== Warmup Phase Performance Metrics With 32 Threads ===

Messages sent: 32000
Successful Message: 32000
Failed Messages: 0
Duration: 4587 ms
Throughput: 6976.24 msg/sec
Success rate: 100.00%

=== Main Phase Performance Metrics With 16 Threads ===

Messages sent: 468000
Successful Message: 468000
Failed Messages: 0
Duration: 28558 ms
Throughput: 16387.70 msg/sec
Success rate: 100.00%

=== Overall Performance Metrics With 16 Threads ===

Messages sent: 500000
Successful Message: 500000
Failed Messages: 0
Duration: 33280 ms
Throughput: 15024.04 msg/sec
Success rate: 100.00%
Total Connections: 20
Total Reconnections: 0
Generating detailed statistical analysis...
Processing CSV data...

Metrics automatically pulled after test: all 500,000 written to DB (see last line in screenshot)

Name: Aijing Huang
CS6650: Scalable Distributed Systems – Assignment 3

```
=====
Fetching metrics from: http://35.94.24.99:8081/metrics
=====
[METRICS] Response:
{
  "messageStats": {
    "total_messages": 500000,
    "earliest_timestamp": "2026-03-31T20:04:27.675436169Z",
    "latest_timestamp": "2026-03-31T20:04:59.338168525Z",
    "messages_per_sec": 15791,
    "messages_per_min": 947478,
    "duration_ms": 31663
  },
  "topUsers": [
    {
      "user_id": "55555",
      "message_count": 17
    },
    {
      "user_id": "71826",
      "message_count": 17
    },
    {
      "user_id": "67476",
      "message_count": 16
    },
    {
      "user_id": "97690",
      "message_count": 16
    },
    {
      "user_id": "15357",
      "message_count": 16
    }
  ],
  "topRooms": [
    {
      "room_id": "6",
      "message_count": 25261
    },
    {
      "room_id": "5",
      "message_count": 25252
    },
    {
      "room_id": "7",
      "message_count": 25222
    },
    {
      "room_id": "20",
      "message_count": 25184
    },
    {
      "room_id": "8",
      "message_count": 25151
    }
  ],
  "userParticipationPatterns": [
    {
      "user_id": "85518",
      "rooms_count": 13,
      "total_messages": 13
    },
    {
      "user_id": "27539",
      "rooms_count": 13,
      "total_messages": 15
    },
    {
      "user_id": "33664",
      "rooms_count": 12,
      "total_messages": 14
    },
    {
      "user_id": "2402",
      "rooms_count": 12,
      "total_messages": 12
    },
    {
      "user_id": "1285",
      "rooms_count": 12,
      "total_messages": 13
    }
  ],
  "totalMessages": 500000
}
```

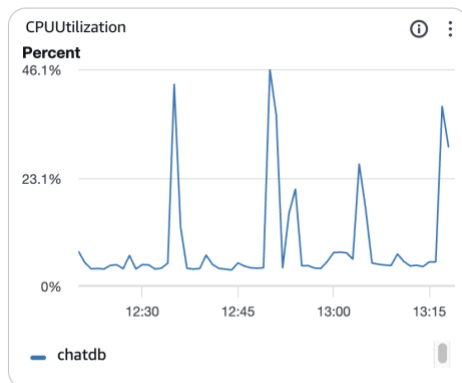
Write throughput, Write latency, Queue Depth

Queue depth: less than max queue size but seems piling up during the test

```
[DB METRICS] dbThroughput=0/sec totalWritten=0 queueDepth=0 p50=0ms p95=0ms p99=0ms
[DB METRICS] dbThroughput=1756/sec totalWritten=8781 queueDepth=1840 p50=53ms p95=95ms p99=113ms
[DB METRICS] dbThroughput=15098/sec totalWritten=84272 queueDepth=2923 p50=30ms p95=51ms p99=56ms
[DB METRICS] dbThroughput=14400/sec totalWritten=156272 queueDepth=14504 p50=33ms p95=47ms p99=57ms
[DB METRICS] dbThroughput=13000/sec totalWritten=221272 queueDepth=31923 p50=33ms p95=65ms p99=97ms
[DB METRICS] dbThroughput=12800/sec totalWritten=285272 queueDepth=49642 p50=32ms p95=74ms p99=133ms
[DB METRICS] dbThroughput=15600/sec totalWritten=363272 queueDepth=54556 p50=30ms p95=41ms p99=61ms
[DB METRICS] dbThroughput=13600/sec totalWritten=431272 queueDepth=68228 p50=31ms p95=74ms p99=127ms
[DB METRICS] dbThroughput=4600/sec totalWritten=454272 queueDepth=45228 p50=70ms p95=338ms p99=599ms
[DB METRICS] dbThroughput=9145/sec totalWritten=500000 queueDepth=0 p50=37ms p95=65ms p99=94ms
[DB METRICS] dbThroughput=0/sec totalWritten=500000 queueDepth=0 p50=0ms p95=0ms p99=0ms
```

This Test was run around 13:05

Database CPU



Database Memory: since RDS provides data for freeableMemory and the freeable memory barely change during the test run, the test barely use any memory.

